Ticket Number	:	BF250917-0001
Date & Time	:	17/09/2025 @ 09:00:21 AM
Analyst Name	:	Heong Wai Kit
Severity Level	:	High
Detection Source	:	linux-alert log
Status	:	True Positive

Executive Summary

An alert was triggered for a possible brute force attack. Investigation was carried out to determine if the activities should be considered suspicious. Brute force attempt was confirmed by john.smith. Privilege was escalated to root to create new user for persistence.

Target Host: tryhackme-2404
Source IP: 10[.]10[.]242[.]248
Current Status: Still Active

Technical Investigation & Timeline

Timeline

17/09/2025 09:01:50.282 AM | Large amount of failed password login by john.smith from 10[.]10[.]242[.]248. – **Brute force attempt**

17/09/2025 09:06:01.591 AM| First accepted password login by john.smith from 10[.]10[.]242[.]248. – **Brute force successful**

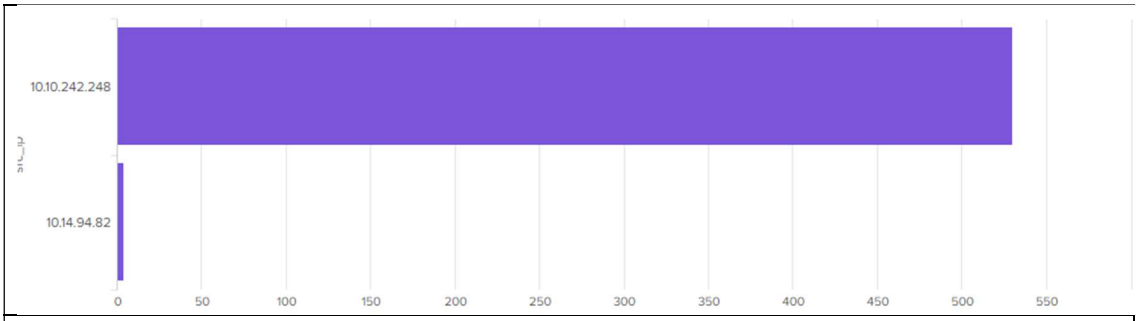
17/09/2025 09:10:03.160 AM | User privilege escalation to root. – **Privilege escalation**

17/09/2025 09:12:10:914 AM | New username created. – **Persistence**

Indicators of Compromise (IOC)

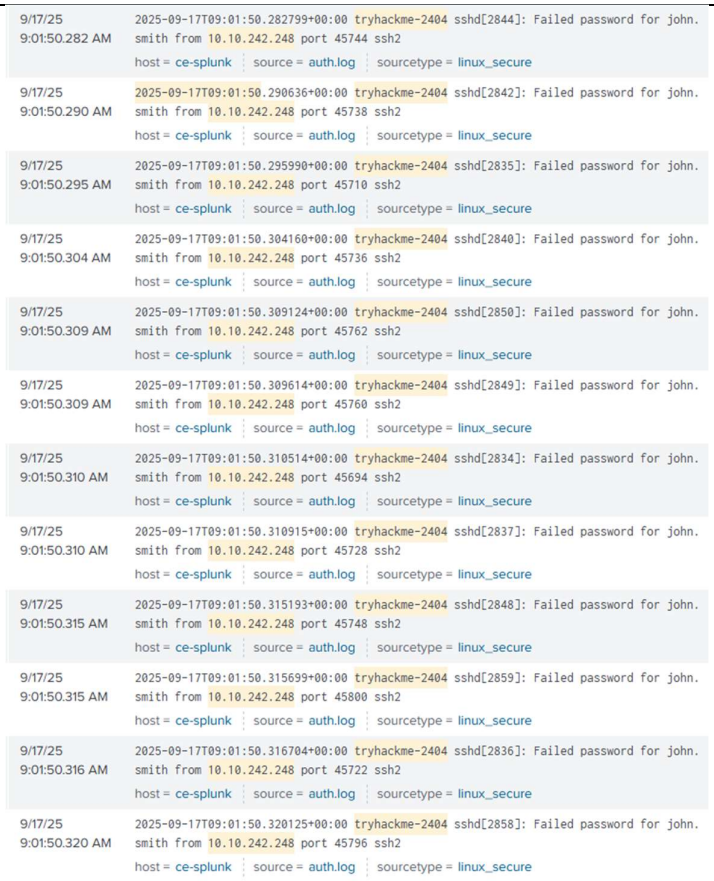
- 500 failed login attempts to by john.smith within 5 minutes.
- Success login by john.smith after high amount of failed login attempts
- Privilege escalation to root.
- New username was created for persistence.

Evidence & Artifacts



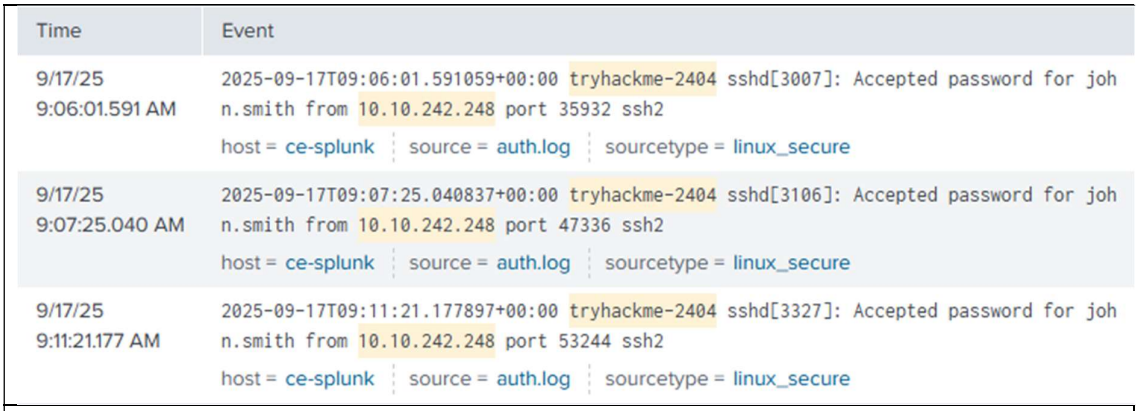
index = "linux-alert" "tryhackme-2404" src_ip!=" "
| table user_name src_ip process
| chart count by src_ip

Large number of login attempts from 10[.]10[.]242[.]248 was recorded.



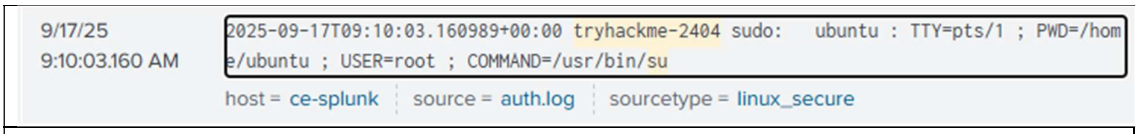
index = "linux-alert" "tryhackme-2404" "10[.]10[.]242[.]248" src_ip!=" "
| sort +_time

Large number of failed password login attempts by john.smith from 10[.]10[.]242[.]248 are recorded.



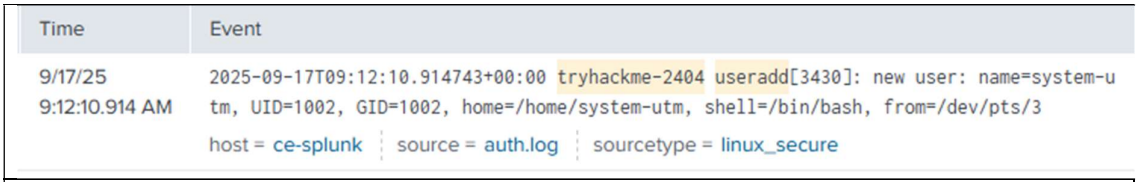
Time	Event
9/17/25 9:06:01.591 AM	2025-09-17T09:06:01.591059+00:00 tryhackme-2404 sshd[3007]: Accepted password for john.smith from 10.10.242.248 port 35932 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
9/17/25 9:07:25.040 AM	2025-09-17T09:07:25.040837+00:00 tryhackme-2404 sshd[3106]: Accepted password for john.smith from 10.10.242.248 port 47336 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
9/17/25 9:11:21.177 AM	2025-09-17T09:11:21.177897+00:00 tryhackme-2404 sshd[3327]: Accepted password for john.smith from 10.10.242.248 port 53244 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure

index = "linux-alert" "tryhackme-2404" "10[.]10[.]242[.]248" src_ip!=" " signature="Accepted password" | sort +_time
Accepted password login by John.smith from 10[.]10[.]242[.]248.



9/17/25 9:10:03.160 AM	2025-09-17T09:10:03.160989+00:00 tryhackme-2404 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/su host = ce-splunk source = auth.log sourcetype = linux_secure

index = "linux-alert" "tryhackme-2404" "su"
Privilage escalation to “root”.



Time	Event
9/17/25 9:12:10.914 AM	2025-09-17T09:12:10.914743+00:00 tryhackme-2404 useradd[3430]: new user: name=system-utm, UID=1002, GID=1002, home=/home/system-utm, shell=/bin/bash, from=/dev/pts/3 host = ce-splunk source = auth.log sourcetype = linux_secure

index = "linux-alert" "tryhackme-2404" "useradd"
Added new username “system-utm”.

Containment & Remediation

Immediate Actions Taken

Escalated to SOC Analyst L2.

Prevention Recommendation

MITRE ATT&CK T1110.001 Brute Force: Password Guessing where the adversary uses systematically guess password.

MITRE D3FEND suggested to implement Multi-factor Authentication and usage of Strong Password Policy to harden.